RAHUL R – 192412255

ECA0502 EMF

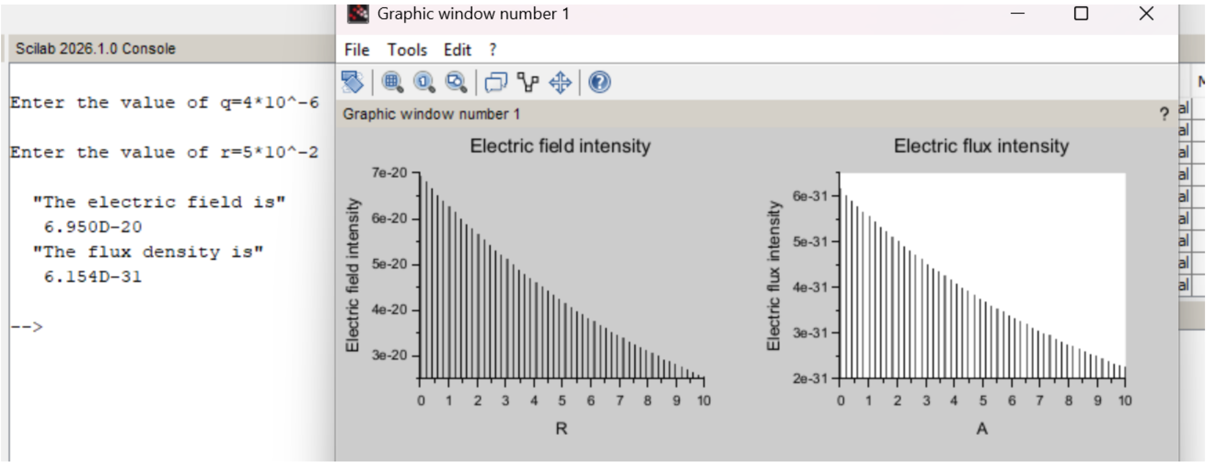
EXP 2

Determination and plotting of Electric field and Electric flux density in sphere full of charges with varying radius.

PROGRAM:

```
clc;
clear;

q=input('Enter the value of q=');
r=input('Enter the value of r=');

efcilon=8.854*10^-12;
pi=3.14
E=q/4*pi*efcilon*r^2;
disp("The electric field is", [E]);
D=E*efcilon;
disp("The flux density is", [D]);
x= linspace(0, 10,50);
y=E*exp(-0.1*x);
z=D* exp(-0.1*x);

figure();
subplot(2, 2,1);
plot2d3(x,y);
xlabel("R");
ylabel("Electric field intensity");
title('Electric field intensity');
subplot(2, 2, 2);
plot2d3(x,z);
xlabel("A");
ylabel("Electric flux intensity");
title('Electric flux intensity');
```

OUTPUT:



OBSERVATION:

EXP-2 Electric field & Electric flux density

i) $q = 4 \times 10^{-6}$ ~~case(i)~~

$\varepsilon_0 = 8.854 \times 10^{-12}$ F/m

ii) $r = 5$ cm

$E = \frac{q}{4\pi\varepsilon_0 r^2}$

$E = \frac{4\times10^{-6}}{2.7795\times10^{-13}}$

$\boxed{E = 6.950\times10^{-20}}$

~~$E = 1.2484\times10^{7}$ V/m~~

O/P ✓

$D = \varepsilon_0 E$

$= 8.854\times10^{-12} \times 6.950\times10^{-20}$

$\boxed{D = 6.154\times10^{-31}}$

CASE 1:

PROGRAM:

EMF EX2 (II).sci (C:\Users\rahul\OneDrive\Documents\EMF EX2 (II).sci) - SciNotes

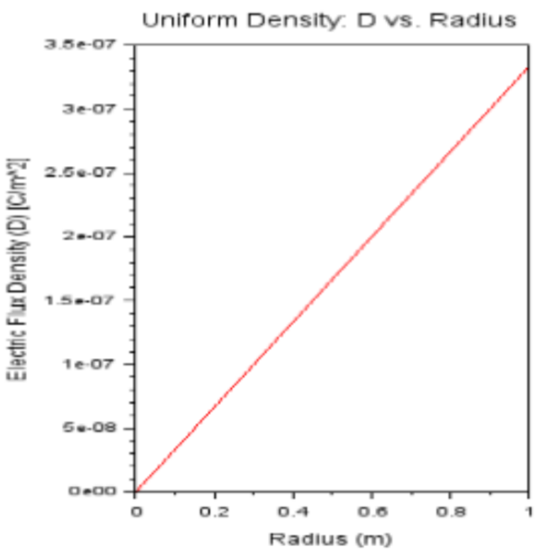
File Edit Format Options Window Execute ?

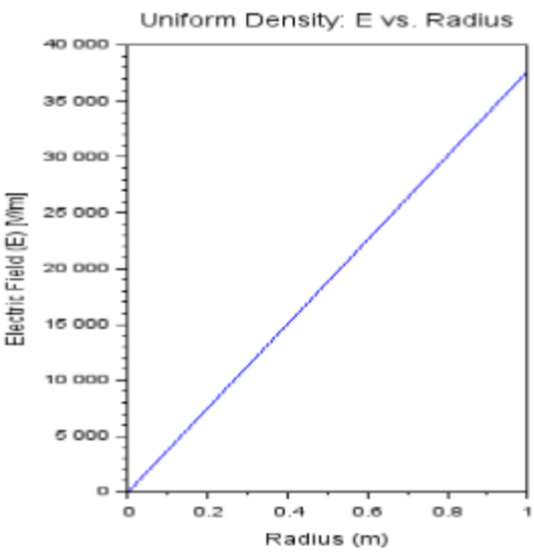
EMF EX2 (I).sci | EMF EX2 (II).sci | EMF EX2 CASE 2.sci | EMF EXP2 CASE 3.sci

```
rho0=1e-6; //Uniform charge density (C/m^3)
epsilon = 8.854e-12; // Permittivity of free space (F/m)
R=1; // Radius of the sphere (m)
n = 100; // Number of steps for plotting
r = linspace(0, R, n); // Radial distances

// Initialize arrays for D and E
D= zeros(1, n);
E= zeros(1, n);
// Calculate D and E
for i= 1:n
ri = r(i); // Current radius
// Enclosed charge Q_enclosed = rho0* (4/3) * pi *r^3
Q_enclosed =rho0*(4/3)*%pi *ri^3;

// Electric Flux Density D
if ri <> 0 then
D(i)=Q_enclosed/(4*%pi*ri^2);
else
D(i) = 0; // Avoid division by zero
end

// Electric Field E = D / epsilon
E(i)= D(i) /epsilon;
end

// Plotting
subplot(1,2,1);
plot(r, D,"r");
xlabel("Radius (m)");
ylabel("Electric Flux Density (D) [C/m^2]");
title("Uniform Density: D vs. Radius");

subplot(1, 2, 2);
```

Line 10, Column 11.

OUTPUT:





CASE 2:

PROGRAM:

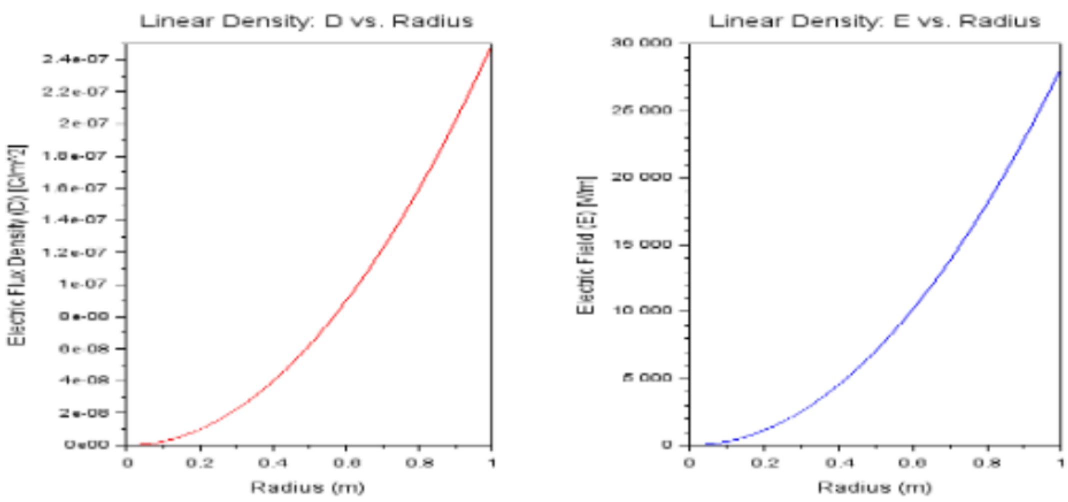
EMF EX2 CASE 2.sci (C:\Users\rahul\OneDrive\Documents\EMF EX2 CASE 2.sci) - SciNotes

File Edit Format Options Window Execute ?

EMF EX2 (I).sci | EMF EX2 (II).sci | EMF EX2 CASE 2.sci | EMF EXP2 CASE 3.sci

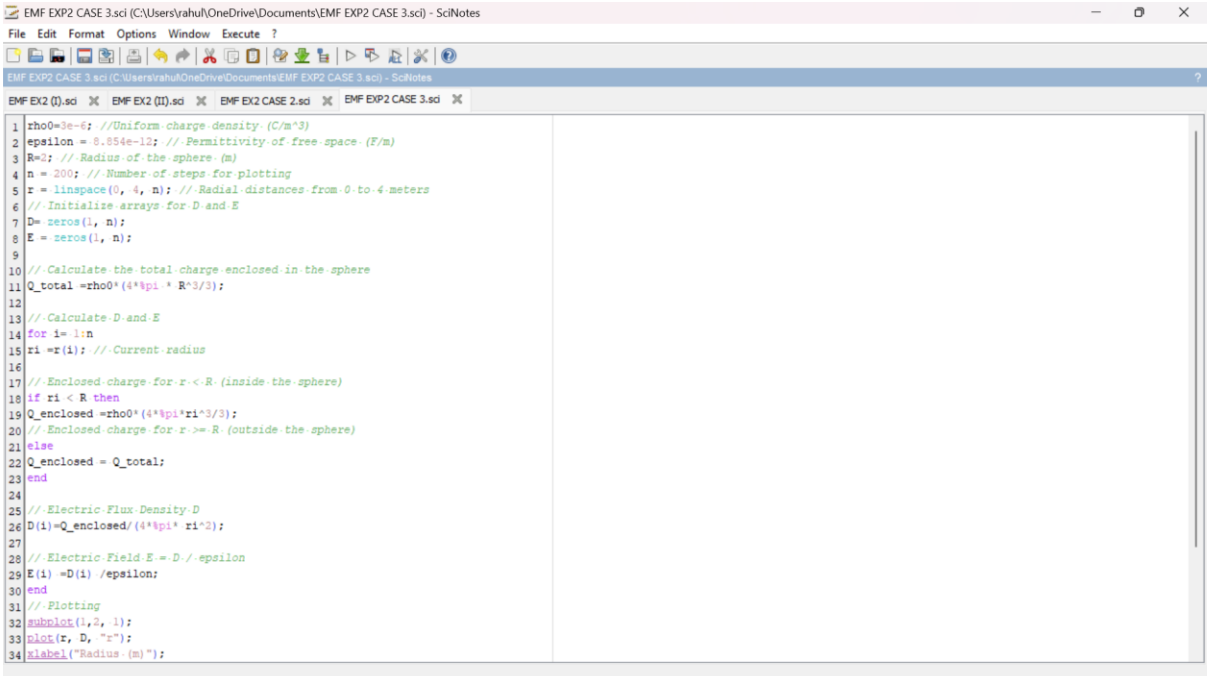
```
n = 100; // Number of steps for plotting
r = linspace(0, R, n); // Radial distances
// Initialize arrays for D and E
D= zeros(1, n);
E= zeros(1, n);
// Calculate D and E
for i = 1:n
ri =r(i); //Current radius
// Enclosed charge Q_enclosed = rho0 *pi *r^4
Q_enclosed =rho0*%pi*ri^4;

// Electric Flux Density D
if ri ~= 0 then
D(i)=Q_enclosed /(4*%pi * ri^2);
else
D(i) = 0; // Avoid division by zero
end

// Electric Field E = D / epsilon
E(i) =D(i)/ epsilon;
end
// Plotting
subplot(1,2,1);
plot(r, D, "r");
xlabel("Radius (m)");
ylabel("Electric Flux Density (D) [C/m^2]");
title("Linear Density: D vs. Radius");

subplot(1,2,2);
plot(r, E, "b");
xlabel("Radius (m)");
ylabel("Electric Field (E) [V/m]");
title("Linear Density: E vs. Radius");
```

OUTPUT:



CASE 3:

PROGRAM:



EMF EXP2 CASE 3.sci (C:\Users\rahul\OneDrive\Documents\EMF EXP2 CASE 3.sci) - SciNotes

File Edit Format Options Window Execute ?

EMF EX2 (I).sci | EMF EX2 (II).sci | EMF EX2 CASE 2.sci | EMF EXP2 CASE 3.sci

```
rho0=3e-6; //Uniform charge density (C/m^3)
epsilon = 8.854e-12; // Permittivity of free space (F/m)
R=2; // Radius of the sphere (m)
n = 200; // Number of steps for plotting
r = linspace(0, 4, n); // Radial distances from 0 to 4 meters
// Initialize arrays for D and E
D= zeros(1, n);
E = zeros(1, n);

// Calculate the total charge enclosed in the sphere
Q_total =rho0*(4*%pi * R^3/3);

// Calculate D and E
for i= 1:n
ri =r(i); // Current radius

// Enclosed charge for r < R (inside the sphere)
if ri < R then
Q_enclosed =rho0*(4*%pi*ri^3/3);
// Enclosed charge for r >= R (outside the sphere)
else
Q_enclosed = Q_total;
end

// Electric Flux Density D
D(i)=Q_enclosed/(4*%pi* ri^2);

// Electric Field E = D / epsilon
E(i) =D(i) /epsilon;
end
// Plotting
subplot(1,2, 1);
plot(r, D, "r");
xlabel("Radius (m)");
```

OUTPUT: